

BSc Semester Assignment Presentation

Ethical Challenges in Open-Source Software Development

HUM 4441: Engineering Ethics

Group Number – 04



GROUP MEMBERS



Mohammad Sadman
Saad

220041115

Syed Md. Nafis Sameen

220041150

Alfi Shahrin

220041153

Asif Hasan

220041156

Introduction

Overview

- Open-source software fosters transparency, collaboration, and innovation.
- Projects like Firefox embody these values, but OSS also faces ethical challenges in misuse, accountability, privacy, and fair competition.

Importance of Ethics

- As software shapes daily life, ethical conduct in IT is vital.
- Unethical design can harm privacy, autonomy, and trust.
- Ethics must go hand-in-hand with performance.

Relevance

- Ethical software practices impact social justice, economic access, and digital freedom.
- As a privacy-first open-source browser, Firefox highlights how ethics can be upheld in large tech projects.

Problem Statement



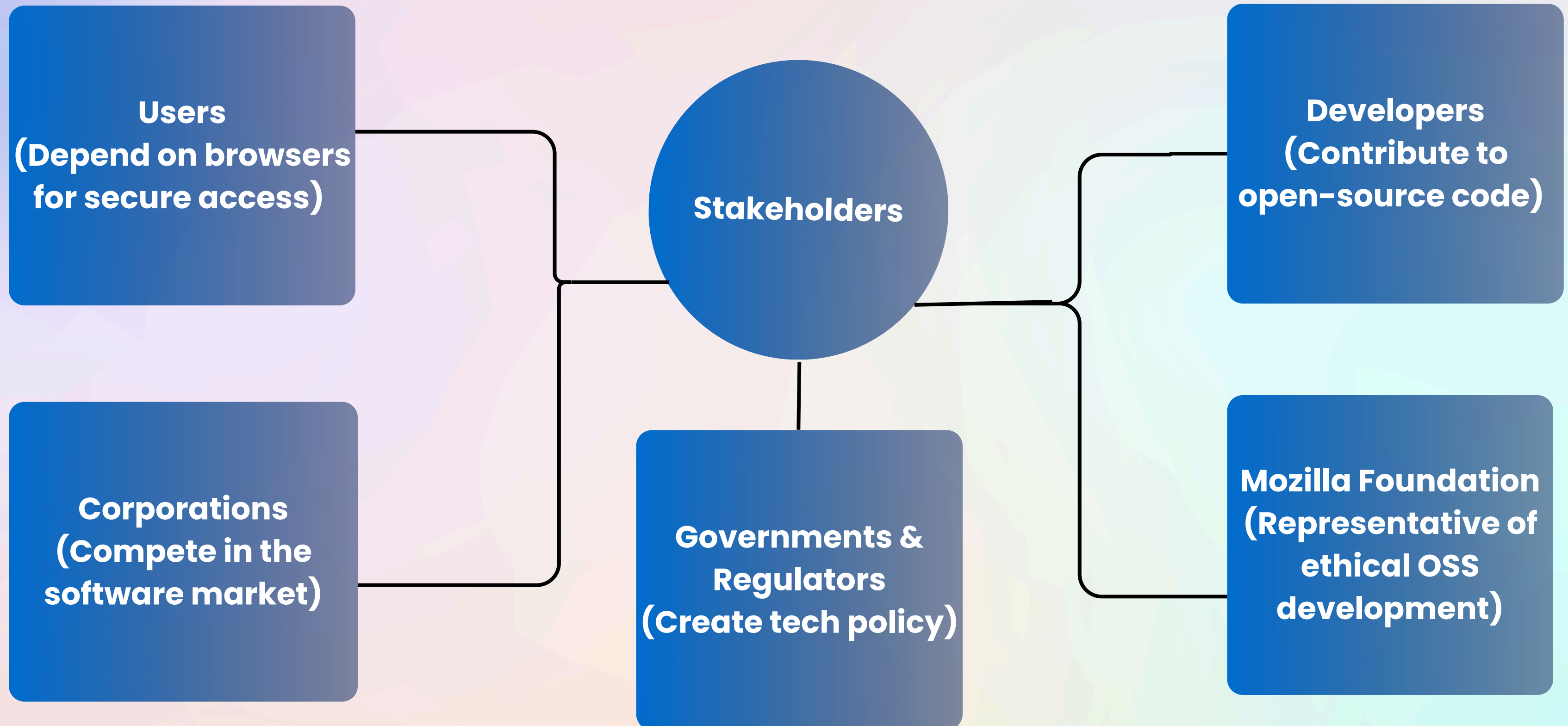
What is the Ethical Issue?

- Open-source software, though built for openness, can be misused or overshadowed by unethical industry dominance.
- Dual-use risks, weak regulation, and disadvantages for ethical options like Firefox are central concerns.

Why is it Important in IT?

- These issues affect user rights, data security, market diversity, and transparency.
- In a time of surveillance and corporate dominance, a browser's ethics are key to preserving user control over their digital identity.

Stakeholders



Objectives



Understand the Ethical Challenges

- Aims to explore the specific ethical tensions that arise in open-source software, especially regarding misuse, accountability, and competition.

Explore Real-World Examples

- Mozilla Firefox is used as the case study to ground theoretical discussions in a real-world setting, illustrating how ethics are applied or challenged in practice.

Recommend Ethical Guidelines

- Based on the findings, our paper will propose practical and policy-level recommendations to support ethical integrity in open-source development and counter unethical behaviours in the broader industry.

Literature Overview

Summary of Existing Research

- Studies highlight the ethical impact of open-source software, with the Mozilla Manifesto promoting openness, transparency, and user control, which sets Firefox apart from profit-driven browsers.
- Barth et al. (2011) warn that extreme transparency (e.g., early patch releases) can risk security, showing ethical trade-offs.
- Comparisons with Chrome and Edge reveal how monopolies can hinder fair, ethical innovation.
- VisionMobile study ranks Firefox moderately in governance openness, showing potential for deeper community involvement.

Ethical Codes

- **IEEE:** Promote public welfare and responsible tech use.
- **ACM:** Avoid harm, respect privacy, be honest.
- **NSPE:** Emphasize public safety and accountability, crucial for software developers.

Similar Cases

- **Tor:** Protects privacy but risks criminal misuse.
- **Chrome:** Feature-rich but known for data tracking and monopoly tactics.
- **Linux & Apache:** Ethical OSS models, but face governance and misuse challenges.

Methodology

Research Methods Used

A qualitative case study of Mozilla Firefox using academic literature, official documentation, community forums, and tech policy reports.

Frameworks for Ethical Analysis

- Utilitarianism (user and societal outcomes)
- Deontology (duties and rights of developers and users)

Approach to Case Study Evaluation

- Governance and transparency
- User privacy protections
- Responses to misuse
- Market impact and regulatory stance

Comparative Analysis

Ethical Criteria	Mozilla Firefox (Open Source)	Chrome/Edge/Safari (Proprietary)
License Type	Open Source (MPL 2.0)	Proprietary, closed-source
Governance	Community + Mozilla Foundation	Centralized corporate control
User Privacy	Tracking protection by default	Data collection common (e.g., Google telemetry)
Transparency	Public development roadmap, open code	Limited insight into internal decisions
Monetization Model	Funded by donations + ethical ad revenue	Ad-based, data-driven, product ecosystem lock-in
Market Share	~3%	Chrome ~65%, Safari ~20%, Edge ~5%

Ethical issues & Analysis



Ethical Challenges in Firefox's Case

- **Licensing & Software Freedom**

Firefox uses the MPL 2.0 “weak copyleft” license, balancing openness with required disclosure of improvements.

- **Community Governance**

While Mozilla promotes open contribution, decisions like a data policy update without input have raised concerns about accountability.

- **Dual-Use Risks**

Privacy tools and open-source access protect users but can also be misused by malicious actors.

- **Market vs. Principles**

With only ~2–3% market share, Firefox relies on search engine deals, raising concerns about the sustainability of its user-first values.

Solutions & Recommendations

- **Technical Improvements**

- Enable stronger default privacy (fingerprinting protection)
- Real-time alerts for harmful extensions
- More user customization
- Launch a lightweight Firefox Lite for low-resource regions

- **Policy Advocacy**

- Push for fair competition (ban browser bundling)
- Promote open-source in education
- Advocate for transparency in data & algorithms

- **Ethics by Design**

- Build in consent-based settings & clear privacy dashboards
- Inform users about data use
- Provide ethical design tools for contributors

Role of Engineers



- **Ethical Principles in Practice**
 - Engineers must uphold Mozilla's values by building features that respect user autonomy.
- **Professional Codes**
 - Guided by ACM/IEEE and NSPE ethics codes, developers aim to avoid harm, protect privacy, and ensure public safety, reviewing features for risks and rejecting unethical changes.
- **Ethical Design**
 - Mozilla's open governance and privacy-by-default features (blocking third-party cookies) show a commitment to transparency and user protection.
- **Innovation vs. Safety**
 - Engineers balance progress with security, ensuring new features are thoroughly tested and do not compromise user data or device safety.

Challenges in implementation

- **Open-Source Governance**

Without central enforcement, ethical compliance depends on community norms, risking misuse and diffuse accountability.

- **Resource Constraints**

Firefox's small market share and non-profit model limit funding, making it hard to scale its ethical approach or compete with tech giants.

- **Business vs. Ethics**

Monetization pressures may conflict with privacy values. Engineers must resist features that prioritize profit over user rights.

- **Emerging Tech Risks**

AI features may introduce hidden tracking or bias. Firefox must ensure transparency, fairness, and user consent in future tech.

- **Regulatory Gaps**

Inconsistent global laws make it hard to uphold a unified ethical standard worldwide.

Future Directions



- **Ethics by Design**
 - Privacy and security are now built-in norms, with laws like GDPR driving features like transparent dashboards and easy tracking opt-outs.
- **Interdisciplinary Collaboration**
 - Ethical tech needs input from engineers, ethicists, legal experts, and user advocates, especially during early design stages.
- **Ethical AI & Cybersecurity**
 - As browsers adopt AI, fairness and auditability are key.
 - Security must evolve to address new threats like quantum or IoT attacks.
- **Data Rights & Trust**
 - Users want control. Firefox can lead by offering tools for data management, clearer access logs, and transparent updates to build trust.

Conclusion



- **Key Findings**

- Despite its strong commitment to user rights, Firefox faces ethical challenges like dual-use risks, governance issues, and market pressures.

- **Importance of Ethics in IT**

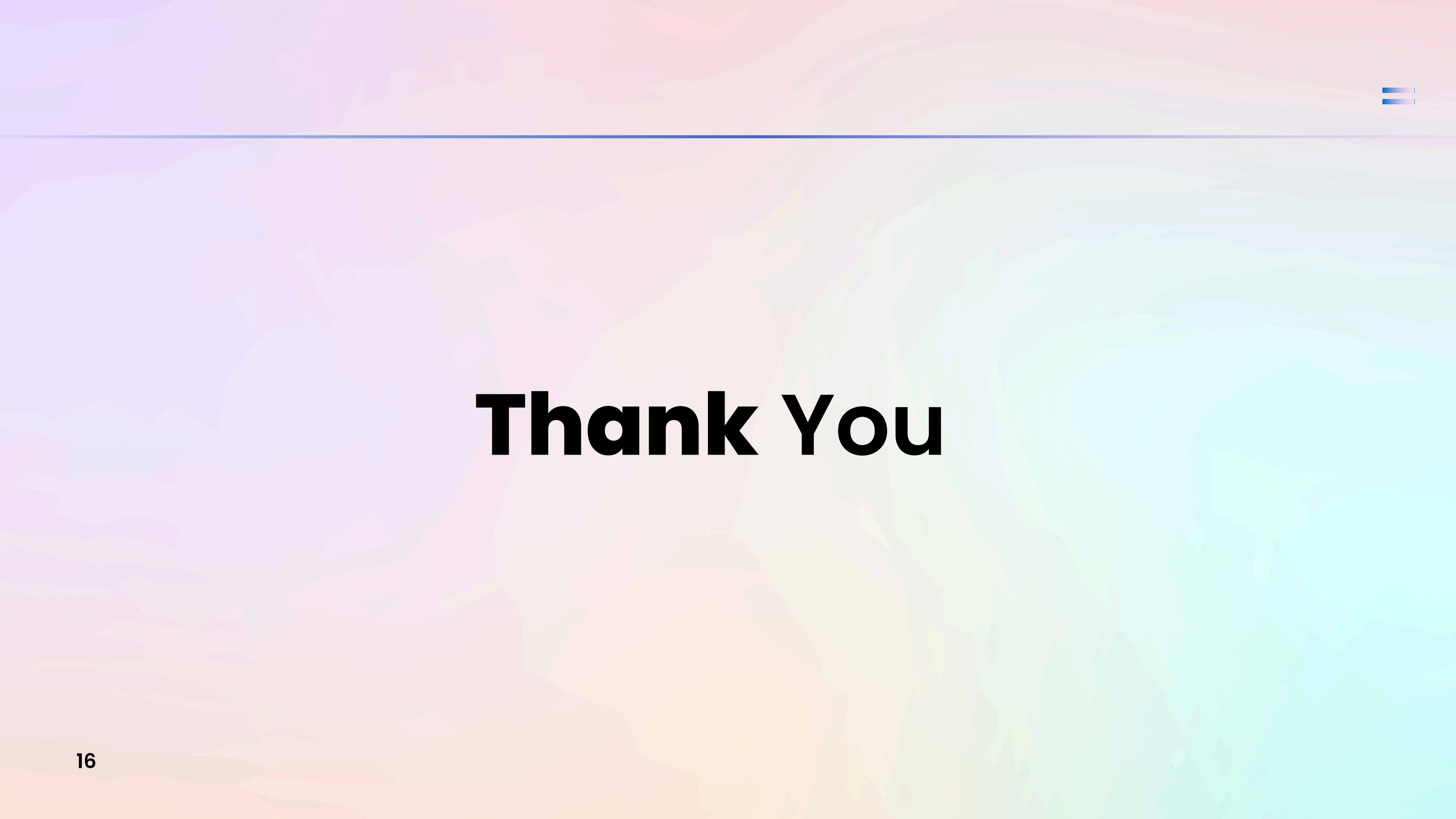
- It proves that privacy and innovation can coexist.
- Ethical IT is essential for protecting freedom and upholding democratic values.

- **Long-Term Implications:**

- Firefox is a model for humane tech, but real impact needs support from users, society, and policy.
- Ethics in tech is a moral choice, not just a technical one.

References

- Mozilla Manifesto (2007/2018)
- VisionMobile Study on Governance Openness
- “How Open Should Open Source Be?” — Barth et al. (2011)
- “Bug-Bounty Program Comparison: Chromium and Firefox” — Ate et al. (2023)
- ACM Code of Ethics (2018)
- IEEE Code of Ethics (2020)
- “Global Browser Market Share” – StatCounter, 2024



Thank You